16 The General Energy Balance Equation

Energy balances are **key** to chemical reactor design. Chemical bonds store **a lot** of energy; that's why breaking and forming bonds in chemical reactions results in large ΔH^{rxn}.

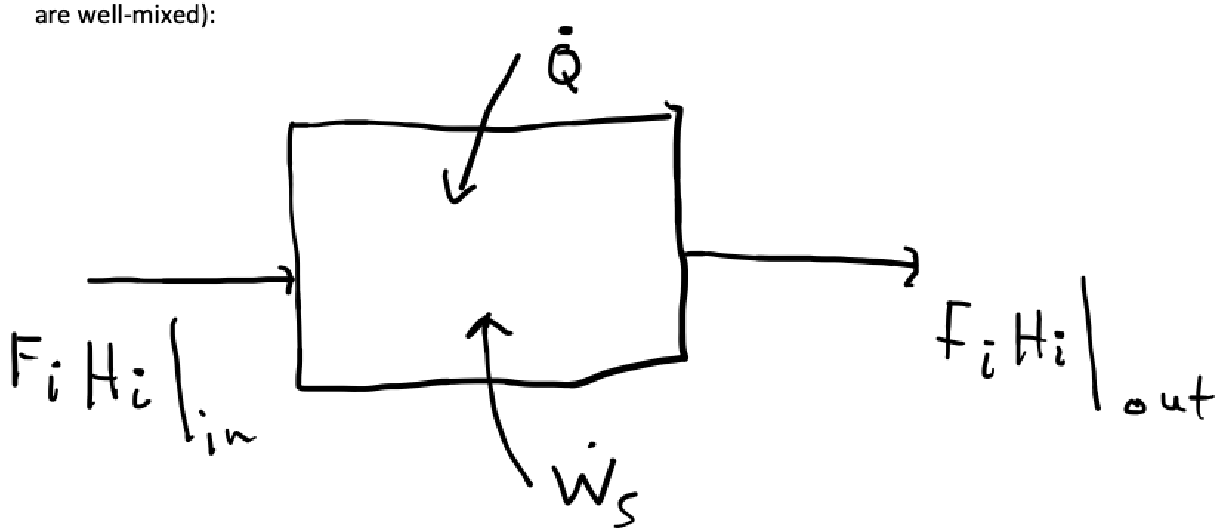
To get a feel for what I mean by large, consider the amount of energy needed to vaporize a mole of water … ~ 40 kJ/mol water. Now consider heat released due to the reaction H_2 (g) + 1/2 O_2 (g) --> H_2O (g) … ~ 240 kJ/mol water.

The reason for the difference is the relative strengths of the bonds being broken and formed: Physical bonds between water molecules in the first case versus chemical bonds between the atoms in a molecule in the second. **Chemical bonds store <u>a lot</u> of potential energy.**

Of course, not all reactions give off as much heat as the H_2 (g) + 1/2 O_2 (g) --> H_2O (g) reaction … The heats of some reactions, e.g., isomerization reactions, are a lot lower, e.g., 40 kJ/mol. However, given the number of chemical species in a reactor, we **need to concern ourselves with all reactions that <u>could</u> happen**, not just those that are desired/designed for.

As we move through this unit on energy balances, we will start by learning how to derive energy balances for common reactors, then we will use the energy balances to study non-isothermal reactor design, then we will examine "reaction runaway," which is often caused when reactions other than those that are desired/designed for occur in a chemical reactor.

To start, consider a system (e.g., a CSTR, or a volume element in a PFR, i.e., something where the contents are well-mixed):



General form of the energy balance:

Rate of accumulation of energy within the system = rate of flow of heat into the system from the surroundings + rate of work done on the system by the surroundings + rate of energy added to the system by mass flow into the system - rate of energy leaving the system by mass flow out of the system.

Symbolically: $\frac{d\hat{E}}{dt} = \dot{Q} + \dot{W}_s + \sum_i F_{i0} H_{i0} - \sum_i F_i H_i$.

The *H*'s are **heats of formation**. E.g., they can be looked up on the NIST website: https://webbook.nist.gov/chemistry/.

Technically, the *H*'s are heats of formation at the reaction temperature. HOWEVER, while the heat of formation for an individual species varies significantly as a function of temperature, ΔH^{rxn} does not vary much at all with reaction temperature. Hence**, it is more important that you obtain a heat of formation for the energy balance than it is to adjust the heat of formation to the reaction temperature**. In fact, a key mistake that students make is to only include the sensible heat changes for each component in their energy balances, instead of including the heat of formation, which includes all of the energy stored within the chemical bonds. Do not make this mistake! Always use the heat of formation, and always check the value that you use against the standard value (which is tabulated).

Key point: You must use the heat of formation in your energy balance; otherwise, you do not account for the energy that is released from the chemical bonds as they break and form.

For a visual on how the energy balance is derived, see the Interactive Computer Games (http://www.umich.edu/~elements/5e/icm/index.html) called "Heat Effects 1" and "Heat Effects 2".

At steady state, the energy balance becomes:

$$0 = \dot{Q} + \dot{W}_s + \sum F_{i0} H_{i0} - \sum F_i H_i$$

Implications on reactor design: The following quantities depend on temperature:

rate constant k: $k(T) = k_1(T_1)\exp\left[\frac{E}{R}\left(\frac{1}{T_1} - \frac{1}{T}\right)\right]$

Concentration for gas phase reactions:

$$C_j = C_{T0}\left(\frac{F_j}{F_T}\right) p \left(\frac{T_0}{T}\right)$$

similarly, T influences v

Equilibrium constant K_p: $K_p(T) = K_{p1}(T_1)\exp\left[\frac{\Delta H^{rxn}}{R}\left(\frac{1}{T_1} - \frac{1}{T}\right)\right]$

Ergun Equation: $\frac{dp}{dW} = -\frac{\alpha}{2p}\frac{T}{T_0}\frac{F_T}{F_{T0}}$

These equations need to be accounted for in non-isothermal reactor design problem solutions.